Options
ID: Fase2_A1G...QAM_TxCanalRx
Title: Fase 2 A...ion optica)
Author: Grupo A1-GI
Description: 16QAM... BER/EVM
Output Language: Python
Generate Options: QT GUI

Complexity: 3.699mbal

Variable
ID: Rs
Value: 3M

Variable
ID: Sps
Value: 8

Variable
ID: samp_rate
Value: *Rs*Sps*=24M

Variable
ID: tabla_de_..._constelacion
Value: *[p/math.…,+3j, 1+1j)]*=[p/math.sqrt(10) fo...

Variable
ID: M
Value: *len(tabl...nstelacion)*=16

Variable
ID: bps
Value: *int(math.log(M,2))*=4

Variable
ID: Rb
Value: *Rs*bps*=12M

Variable
ID: span
Value: 11

Variable
ID: E_pulso
Value: *sum(t*t ...rc_tx_taps)*=7.94135

Variable
ID: rrc_rx_taps
Value: *[t/E_pul...rc_tx_taps]*=[t/E_pulso for t in...

Variable
ID: constelacion_16qam
Value: *digital....MALIZATION)*=<gnuradi...b909305db0>

RRC Filter Taps
ID: rrc_tx_taps
Gain: *Sps*=8
Sample Rate (Hz): *samp_rate*=24M
Symbol Rate (Hz): *Rs*=3M
Excess BW: *beta*=350m
Num Taps: *span*Sps*=88

Variable
ID: noise_amp
Value: *awgn_on*...sN0_dB/10))*=89.1143m

Variable
ID: g_sombra
Value: *10**(-atten_dB/20)*=1

Variable
ID: emi_rate
Value: *16e3*=16k

Variable
ID: emi_dur
Value: *1e-6*=1u

Variable
ID: emi_amp
Value: *math.sqr...isr_dB/10))*=996.328m

Variable
ID: ref_agc
Value: *sum(abs(...telacion)/M*=947.214m

Variable
ID: agc_rate
Value: *1e-3*=1m

Variable
ID: ber_win
Value: 200k

Variable
ID: modo_captura
Value: 0

Import
ID: import_0
Import: *import ...port firdes*=math, digital, firdes

Python Snippet
ID: captura_auto
Section of Flowgraph: Main - After Start
Code Snippet: # --...art(50)

Guion automatico de pruebas

QT GUI Range
ID: EsN0_dB
Label: Es/N0 LOS [dB]
Default Value: 30
Start: 0
Stop: 45
Step: 500m

QT GUI Range
ID: atten_dB
Label: Atenuaci...sombra [dB]
Default Value: 0
Start: 0
Stop: 25
Step: 1

QT GUI Range
ID: emi_isr_dB
Label: EMI: pot.../senal [dB]
Default Value: 0
Start: -15
Stop: 10
Step: 1

QT GUI Range
ID: d_muestreo
Label: Retardo ... [muestras]
Default Value: 7
Start: 0
Stop: 7
Step: 1

QT GUI Range
ID: retardo_ref
Label: Retardo ... [simbolos]
Default Value: 11
Start: 0
Stop: 30
Step: 1

QT GUI Range
ID: beta
Label: Roll-off beta
Default Value: 350m
Start: 100m
Stop: 1
Step: 50m

QT GUI Check Box
ID: awgn_on
Label: AWGN de fondo
Default Value: 1
True: 1
False: 0

QT GUI Check Box
ID: emi_on
Label: EMI impu.../soldadura)
Default Value: 0
True: 1
False: 0

QT GUI Check Box
ID: agc_on
Label: AGC en el receptor
Default Value: 1
True: 1
False: 0

Random Source
ID: fuente_bits
Minimum: 0
Maximum: 2
Num Samples: 131.072k
Repeat: Yes

Bits aleatorios (2^17, periodo largo)

Repack Bits
ID: bits_a_simbolos
Bits per input byte: 1
Bits per output byte: *bps*=4

Agrupa 4 bits → indice 0..15

Chunks to Symbols
ID: mapeo_16qam
Symbol Table: *tabl...telacion*=tabla_de_ver...
Dimension: 1

Mapeo Gray

Interpolating FIR Filter
ID: rrc_tx
Interpolation: *Sps*=8
Taps: *rrc_tx_taps*=rrc_tx_taps

Conformador RRC Tx

Throttle
ID: throttle_0
Sample Rate: *samp_rate*=24M
Limit: None

Multiply Const
ID: canal_sombra
Constant: *g_sombra*=1

Sombra: atenuacion de la senal

Add
ID: canal_suma

QT GUI Frequency Sink
ID: psd_tx_canal
Name: *"PSD: sa...a del canal"*=PSD: sal...da del canal
FFT Size: 2048
Center Frequency (Hz): 0
Bandwidth (Hz): *samp_rate*=24M

QT GUI Frequency Sink
ID: psd_beta
Name: *"PSD Tx ...os roll-off"*=PSD Tx p...tos roll-off
FFT Size: 2048
Center Frequency (Hz): 0
Bandwidth (Hz): *samp_rate*=24M

Interpolating FIR Filter
ID: rrc_b020
Interpolation: *Sps*=8
Taps: *firdes.r...0, span*Sps)*=firdes.root_raised_c...

Noise Source
ID: ruido_awgn
Noise Type: *analo..._GAUSSIAN*=Gaussian
Amplitude: *noise_amp*=89.1143m
Seed: 0

Ruido de fondo

Interpolating FIR Filter
ID: rrc_b050
Interpolation: *Sps*=8
Taps: *firdes.r...0, span*Sps)*=firdes.root_raised_c...

beta = 0.50 (comparacion)

Ruido impulsivo EMI
ID: ruido_emi
Samp_Rate: *samp_rate*=24M
Rate: *emi_rate*=16k
Dur: *emi_dur*=1u
Amp: *emi_amp*=996.328m
Enable: *emi_on*=0

EMI impulsiva Bernoulli-Gaussiana

QT GUI Time Sink
ID: envolvente_canal
Name: *"Envolve...a del canal"*=Envolven...da del canal
Number of Points: 4.096k
Sample Rate: *samp_rate*=24M
Autoscale: No

QT GUI Constellation Sink
ID: const_canal
Name: *"Plano I... del canal)"*=Plano I-...a del canal)
Number of Points: 4.096k
Autoscale: No

Decimating FIR Filter
ID: rx_filtro_acoplado
Decimation: 1
Taps: *rrc_rx_taps*=rrc_rx_taps

Filtro acoplado RRC

Delay
ID: rx_ajuste_muestreo
Delay: *d_muestreo*=7

Fase de muestreo

Keep 1 in N
ID: rx_muestreo

1 muestra por simbolo

AGC
ID: rx_agc
Rate: *agc_rate*agc_on*=1m
Reference: *ref_agc*=947.214m
Gain: 1
Max Gain: 100

Compensa la atenuacion (sombra)

Constellation Decoder
ID: rx_decisor
Constellation Object: *...6qam=...9f0>*

Minima distancia

Chunks to Symbols
ID: rx_punto_ideal
Symbol Table: *tabl...telacion*=tabla_de_ver...
Dimension: 1

Subtract
ID: evm_error

Complex to Mag^2
ID: evm_mag2

Moving Average
ID: evm_promedio
Length: 20k
Scale: *1/20000*=50u
Max Iter: 4k
Length of Vectors: 1

Transcendental
ID: evm_raiz
Function Name: sqrt

Multiply Const
ID: evm_pct
Constant: 100

Probe Signal
ID: probe_evm

QT GUI Eye Sink
ID: ojo_filtro_acoplado
Number of Points: 1.024k
Samples per Symbol: *Sps*=8
Sample Rate: *samp_rate*=24M
Autoscale: No

QT GUI Constellation Sink
ID: const_rx
Name: *"Constel...ra/simbolo)"*=Constela...tra/simbolo)
Number of Points: 2.048k
Autoscale: No

Delay
ID: ref_retardo
Delay: *retardo_ref*=11

Alinea la referencia Tx

Xor
ID: ber_xor

Unpack K Bits
ID: ber_unpack
K: *bps*=4

UChar To Float
ID: ber_float

Moving Average
ID: ber_promedio
Length: *ber_win*=200k
Scale: *1.0/ber_win*=5u
Max Iter: 4k
Length of Vectors: 1

Probe Signal
ID: probe_ber

QT GUI Number Sink
ID: evm_num
Name: *"EVM rms"*=EVM rms
Autoscale: No
Average: 0
Graph Type: Horizontal

QT GUI Number Sink
ID: ber_num
Name: *"BER medida"*=BER medida
Autoscale: No
Average: 0
Graph Type: Horizontal

UChar To Float
ID: ref_float

UChar To Float
ID: rx_float

QT GUI Time Sink
ID: simbolos_tx_rx
Name: *"Simbolo...ndice 0-15)"*=Simbolos...indice 0-15)
Number of Points: 48
Sample Rate: *Rs*=3M
Autoscale: No